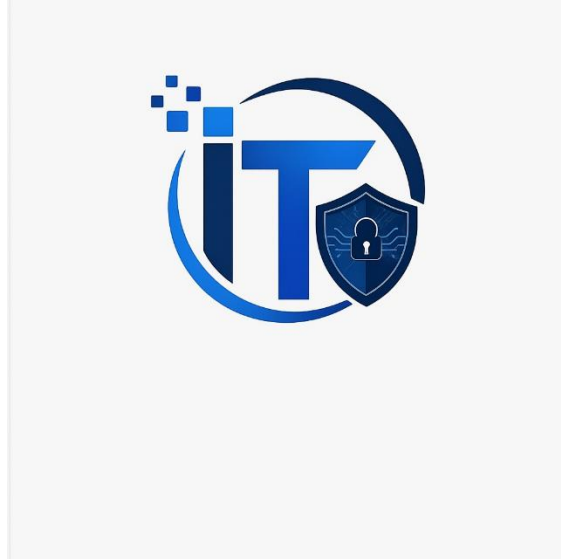


ITSIMPLERA SOLUTIONS

Network Administration Internship Program



WEEK 07 ASSIGNMENT

Enterprise VPN, Firewall, and Secure Access Implementation

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Company: IT-SIMPLERA Institute

Domain: Network Administration

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1. Introduction

This report documents the design and implementation of a secure enterprise network infrastructure using GNS3 simulator. The objective was to establish secure communication between branch offices using IPsec VPNs, protect internal resources using Zone-Based Firewalls (ZBF), and secure network devices through AAA authentication, SSH, device hardening, and centralized logging.

The network consists of three sites: Headquarters (HQ), Branch-A, and Branch-B, all interconnected through an ISP router simulating the internet/WAN environment.

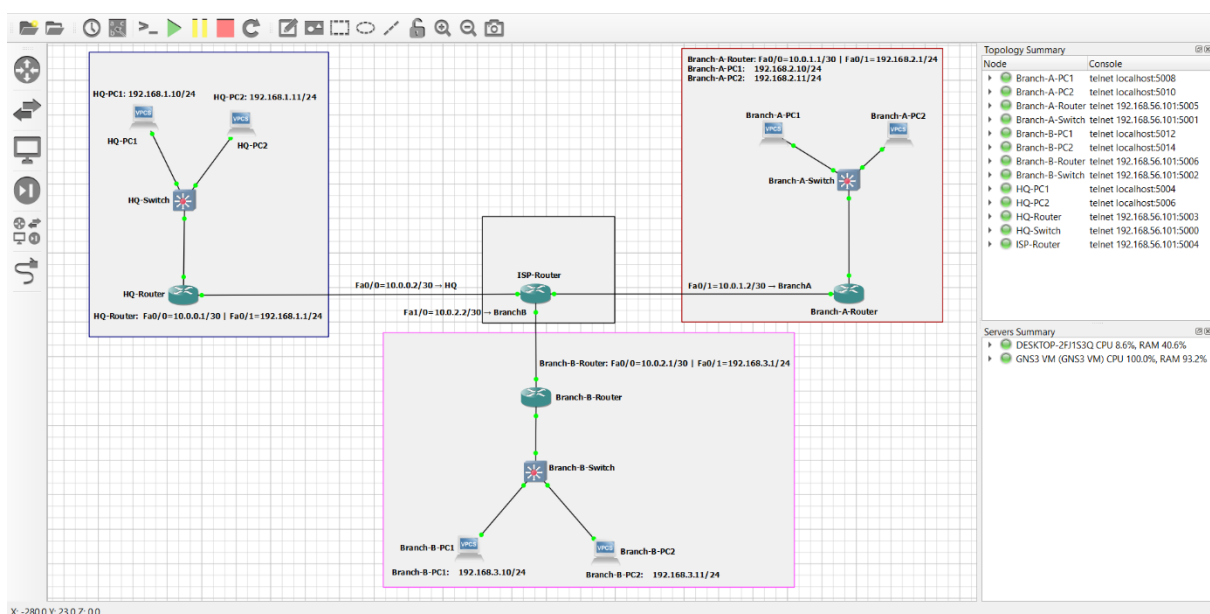
2. Network Topology

The network topology consists of the following devices:

- 4 Routers: HQ-Router, ISP-Router, Branch-A-Router, Branch-B-Router (Cisco c3725)
- 3 Switches: HQ-Switch, Branch-A-Switch, Branch-B-Switch (Cisco IOU L2)
- 6 PCs: HQ-PC1, HQ-PC2, Branch-A-PC1, Branch-A-PC2, Branch-B-PC1, Branch-B-PC2 (VPCS)

Topology Description:

Each branch router connects to the ISP-Router via WAN links (FastEthernet). The ISP-Router acts as the internet/WAN cloud. Each site has a local switch connecting PCs to the router LAN interface. IPsec VPN tunnels are established between HQ and each branch through the ISP.



3. IP Addressing Scheme

WAN Links:

Device	Interface	IP Address	Purpose
HQ-Router	Fa0/0	10.0.0.1/30	WAN to ISP
HQ-Router	Fa0/1	192.168.1.1/24	LAN
ISP-Router	Fa0/0	10.0.0.2/30	WAN to HQ
ISP-Router	Fa0/1	10.0.1.2/30	WAN to BranchA
ISP-Router	Fa1/0	10.0.2.2/30	WAN to BranchB
Branch-A-Router	Fa0/0	10.0.1.1/30	WAN to ISP
Branch-A-Router	Fa0/1	192.168.2.1/24	LAN
Branch-B-Router	Fa0/0	10.0.2.1/30	WAN to ISP
Branch-B-Router	Fa0/1	192.168.3.1/24	LAN

PC IP Addresses:

PC	IP Address	Gateway
HQ-PC1	192.168.1.10/24	192.168.1.1
HQ-PC2	192.168.1.11/24	192.168.1.1
Branch-A-PC1	192.168.2.10/24	192.168.2.1
Branch-A-PC2	192.168.2.11/24	192.168.2.1
Branch-B-PC1	192.168.3.10/24	192.168.3.1
Branch-B-PC2	192.168.3.11/24	192.168.3.1

4. OSPF Routing Configuration

OSPF (Open Shortest Path First) was configured on all routers to enable dynamic routing between all sites. Process ID 1 was used with Area 0 (backbone area). Passive interfaces were configured on LAN interfaces to prevent unnecessary OSPF hello packets from being sent to end devices.

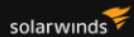
OSPF Router IDs:

- **HQ-Router: 1.1.1.1**

```
HQ-Router#
HQ-Router#en
HQ-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#router ospf 1
HQ-Router(config-router)# router-id 1.1.1.1
HQ-Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
HQ-Router(config-router)# network 10.0.0.0 0.0.0.3 area 0
HQ-Router(config-router)# passive-interface FastEthernet0/1
HQ-Router(config-router)#exit
HQ-Router(config)#end
HQ-Router#write memory
*Mar  1 00:20:39.475: %SYS-5-CONFIG_I: Configured from console by console
HQ-Router#write memory
Building configuration...
[OK]
HQ-Router#
```

- **ISP-Router: 2.2.2.2**

```
ISP-Router#
ISP-Router#en
ISP-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ISP-Router(config)#router ospf 1
ISP-Router(config-router)# router-id 2.2.2.2
ISP-Router(config-router)# network 10.0.0.0 0.0.0.3 area 0
ISP-Router(config-router)# network 10.0.1.0 0.0.0.3 area 0
ISP-Router(config-router)# network 10.0.2.0 0.0.0.3 area 0
ISP-Router(config-router)#exit
ISP-Router(config)#end
ISP-Router#write memory
*Mar 1 00:21:18.115: %SYS-5-CONFIG_I: Configured from console by console
ISP-Router#write memory
*Mar 1 00:21:19.627: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
ISP-Router#write memory
Building configuration...
[OK]
ISP-Router#
```

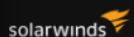


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- **Branch-A-Router: 3.3.3.3**

```
Branch-A-Router#
Branch-A-Router#en
Branch-A-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Branch-A-Router(config)#router ospf 1
Branch-A-Router(config-router)# router-id 3.3.3.3
Branch-A-Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Branch-A-Router(config-router)# network 10.0.1.0 0.0.0.3 area 0
Branch-A-Router(config-router)# passive-interface FastEthernet0/1
Branch-A-Router(config-router)#exit
Branch-A-Router(config)#end
Branch-A-Router#write memory
*Mar 1 00:21:53.875: %SYS-5-CONFIG_I: Configured from console by console
Branch-A-Router#write memory
Building configuration...
[OK]
Branch-A-Router#
*Mar 1 00:21:59.727: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0 from LOADING to FULL, Loading Done
Branch-A-Router#
```



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- **Branch-B-Router: 4.4.4.4**

```
Branch-B-Router#
Branch-B-Router#en
Branch-B-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Branch-B-Router(config)#router ospf 1
Branch-B-Router(config-router)# router-id 4.4.4.4
Branch-B-Router(config-router)# network 192.168.3.0 0.0.0.255 area 0
Branch-B-Router(config-router)# network 10.0.2.0 0.0.0.3 area 0
Branch-B-Router(config-router)# passive-interface FastEthernet0/1
Branch-B-Router(config-router)#exit
Branch-B-Router(config)#end
Branch-B-Router#write memory
*Mar 1 00:22:23.835: %SYS-5-CONFIG_I: Configured from console by console
Branch-B-Router#write memory
Building configuration...
[OK]
Branch-B-Router#
*Mar 1 00:22:30.255: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0 from LOADING to FULL, Loading Done
Branch-B-Router#
```



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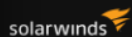
Verification:

```
HQ-Router#
HQ-Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1    FULL/BDR        00:00:35   10.0.0.2       FastEthernet0/0
HQ-Router#ping 192.168.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/121/208 ms
HQ-Router#ping 192.168.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/84/168 ms
HQ-Router#
```



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5. IPsec VPN Configuration

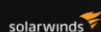
IPsec Site-to-Site VPN tunnels were configured between all three sites to ensure encrypted communication over the public internet. The following technologies were used:

- ISAKMP (IKE Phase 1): AES encryption, SHA hashing, pre-shared keys, DH Group 2
- IPsec (Phase 2): ESP-AES encryption, ESP-SHA-HMAC authentication
- Pre-shared Key: cisco123

VPN Tunnels:

- **Tunnel 1: HQ-Router (10.0.0.1) ↔ Branch-A-Router (10.0.1.1)**

```
HQ-Router(config)#
HQ-Router(config)#crypto isakmp policy 10
HQ-Router(config-isakmp)# encr aes
HQ-Router(config-isakmp)# hash sha
HQ-Router(config-isakmp)# authentication pre-share
HQ-Router(config-isakmp)# group 2
HQ-Router(config-isakmp)# lifetime 86400
HQ-Router(config-isakmp)#
HQ-Router(config-isakmp)#crypto isakmp key cisco123 address 10.0.1.1
HQ-Router(config)#crypto isakmp key cisco123 address 10.0.2.1
HQ-Router(config)#
HQ-Router(config)#crypto ipsec transform-set TSET esp-aes esp-sha-hmac
HQ-Router(cfg-crypto-trans)#
HQ-Router(cfg-crypto-trans)#crypto map CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
HQ-Router(config-crypto-map)# set peer 10.0.1.1
HQ-Router(config-crypto-map)# set transform-set TSET
HQ-Router(config-crypto-map)# match address VPN-ACL-A
HQ-Router(config-crypto-map)#
HQ-Router(config-crypto-map)#crypto map CMAP 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
HQ-Router(config-crypto-map)# set peer 10.0.2.1
HQ-Router(config-crypto-map)# set transform-set TSET
HQ-Router(config-crypto-map)# match address VPN-ACL-B
HQ-Router(config-crypto-map)#
HQ-Router(config-crypto-map)#ip access-list extended VPN-ACL-A
HQ-Router(config-ext-nacl)#192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
HQ-Router(config-ext-nacl)#
HQ-Router(config-ext-nacl)#ip access-list extended VPN-ACL-B
HQ-Router(config-ext-nacl)#192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
HQ-Router(config-ext-nacl)#
HQ-Router(config-ext-nacl)#interface FastEthernet0/0
HQ-Router(config-if)# crypto map CMAP
HQ-Router(config-if)#
HQ-Router(config-if)#end
HQ-Router#write memory
*Mar  1 00:24:45.431: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
HQ-Router#write memory
*Mar  1 00:24:45.571: %SYS-5-CONFIG_I: Configured from console by console
HQ-Router#write memory
```



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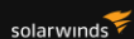
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- **Tunnel 2: HQ-Router (10.0.0.1) ↔ Branch-B-Router (10.0.2.1)**

```
Branch-B-Router(config)#
Branch-B-Router(config)#crypto isakmp policy 10
Branch-B-Router(config-isakmp)# encr aes
Branch-B-Router(config-isakmp)# hash sha
Branch-B-Router(config-isakmp)# authentication pre-share
Branch-B-Router(config-isakmp)# group 2
Branch-B-Router(config-isakmp)# lifetime 86400
Branch-B-Router(config-isakmp)#
Branch-B-Router(config-isakmp)#crypto isakmp key cisco123 address 10.0.0.1
Branch-B-Router(config)#crypto isakmp key cisco123 address 10.0.1.1
Branch-B-Router(config)#
Branch-B-Router(config)#crypto ipsec transform-set TSET esp-aes esp-sha-hmac
Branch-B-Router(cfg-crypto-trans)#
Branch-B-Router(cfg-crypto-trans)#crypto map CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Branch-B-Router(config-crypto-map)# set peer 10.0.0.1
Branch-B-Router(config-crypto-map)# set transform-set TSET
Branch-B-Router(config-crypto-map)# match address VPN-ACL-HQ
Branch-B-Router(config-crypto-map)#
Branch-B-Router(config-crypto-map)#crypto map CMAP 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Branch-B-Router(config-crypto-map)# set peer 10.0.1.1
Branch-B-Router(config-crypto-map)# set transform-set TSET
Branch-B-Router(config-crypto-map)# match address VPN-ACL-A
Branch-B-Router(config-crypto-map)#
Branch-B-Router(config-crypto-map)#ip access-list extended VPN-ACL-HQ
Branch-B-Router(config-ext-nacl)#$192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
Branch-B-Router(config-ext-nacl)#
Branch-B-Router(config-ext-nacl)#ip access-list extended VPN-ACL-A
Branch-B-Router(config-ext-nacl)#$192.168.3.0 0.0.0.255 192.168.2.0 0.0.0.255
Branch-B-Router(config-ext-nacl)#
Branch-B-Router(config-ext-nacl)#interface FastEthernet0/0
Branch-B-Router(config-if)# crypto map CMAP
Branch-B-Router(config-if)#
Branch-B-Router(config-if)#end
Branch-B-Router#write memory
*Mar  1 00:51:56.675: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
Branch-B-Router#write memory
*Mar  1 00:51:56.703: %SYS-5-CONFIG_I: Configured from console by console
Branch-B-Router#write memory
Building configuration...
[OK]
Branch-B-Router#
```

- **Tunnel 3: Branch-A-Router (10.0.1.1) ↔ Branch-B-Router (10.0.2.1)**

```
Branch-A-Router(config)#
Branch-A-Router(config)#crypto isakmp policy 10
Branch-A-Router(config-isakmp)# encr aes
Branch-A-Router(config-isakmp)# hash sha
Branch-A-Router(config-isakmp)# authentication pre-share
Branch-A-Router(config-isakmp)# group 2
Branch-A-Router(config-isakmp)# lifetime 86400
Branch-A-Router(config-isakmp)#
Branch-A-Router(config-isakmp)#crypto isakmp key cisco123 address 10.0.0.1
Branch-A-Router(config)#crypto isakmp key cisco123 address 10.0.2.1
Branch-A-Router(config)#
Branch-A-Router(config)#crypto ipsec transform-set TSET esp-aes esp-sha-hmac
Branch-A-Router(cfg-crypto-trans)#
Branch-A-Router(cfg-crypto-trans)#crypto map CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
      and a valid access list have been configured.
Branch-A-Router(config-crypto-map)# set peer 10.0.0.1
Branch-A-Router(config-crypto-map)# set transform-set TSET
Branch-A-Router(config-crypto-map)# match address VPN-ACL-HQ
Branch-A-Router(config-crypto-map)#
Branch-A-Router(config-crypto-map)#crypto map CMAP 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
      and a valid access list have been configured.
Branch-A-Router(config-crypto-map)# set peer 10.0.2.1
Branch-A-Router(config-crypto-map)# set transform-set TSET
Branch-A-Router(config-crypto-map)# match address VPN-ACL-B
Branch-A-Router(config-crypto-map)#
Branch-A-Router(config-crypto-map)#ip access-list extended VPN-ACL-HQ
Branch-A-Router(config-ext-nacl)#$192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Branch-A-Router(config-ext-nacl)#
Branch-A-Router(config-ext-nacl)#ip access-list extended VPN-ACL-B
Branch-A-Router(config-ext-nacl)#$192.168.2.0 0.0.0.255 192.168.3.0 0.0.0.255
Branch-A-Router(config-ext-nacl)#
Branch-A-Router(config-ext-nacl)#interface FastEthernet0/0
Branch-A-Router(config-if)# crypto map CMAP
Branch-A-Router(config-if)#
Branch-A-Router(config-if)#end
Branch-A-Router#write memory
*Mar  1 00:26:00.663: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
Branch-A-Router#write memory
*Mar  1 00:26:00.831: %SYS-5-CONFIG_I: Configured from console by console
Branch-A-Router#write memory
Building configuration...
[OK]
Branch-A-Router#
```



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Verification:

```
HQ-Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst      src      state    conn-id slot status
10.0.1.1 10.0.0.1 QM_IDLE 1001    0 ACTIVE
10.0.2.1 10.0.0.1 QM_IDLE 1002    0 ACTIVE

IPv6 Crypto ISAKMP SA

HQ-Router#show crypto ipsec sa
Interface: FastEthernet0/0
Crypto map tag: CMAP, local addr 10.0.0.1

protected urf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.1.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
  #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

  local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.1.1
  path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
  current outbound spi: 0x377C6659(939288153)

inbound esp sas:
  spi: 0x675C4BE(108381374)
    transform: esp-aes esp-sha-hmac ,
  --More--
```

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```
HQ-Router#ping 192.168.2.1 source 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/88/108 ms
HQ-Router#ping 192.168.3.1 source 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 76/115/148 ms
HQ-Router#

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HQ-PC1
HQ-PC1> ping 192.168.2.10
192.168.2.10 icmp_seq=1 timeout
192.168.2.10 icmp_seq=2 timeout
64 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=240.557 ms
64 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=249.437 ms
64 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=129.139 ms
HQ-PC1> HQ-PC1> ping 192.168.3.10
Bad command: "HQ-PC1> ping 192.168.3.10". Use ? for help.
HQ-PC1> ping 192.168.3.10
192.168.3.10 icmp_seq=1 timeout
192.168.3.10 icmp_seq=2 timeout
64 bytes from 192.168.3.10 icmp_seq=3 ttl=62 time=146.639 ms
64 bytes from 192.168.3.10 icmp_seq=4 ttl=62 time=137.851 ms
64 bytes from 192.168.3.10 icmp_seq=5 ttl=62 time=150.368 ms
HQ-PC1>
```

6. Zone-Based Firewall (ZBF)

Zone-Based Firewall was configured on all three routers (HQ, Branch-A, Branch-B) to protect internal resources from unauthorized external access. Two security zones were defined:

- INSIDE Zone: LAN interface (FastEthernet0/1) — trusted internal network
- OUTSIDE Zone: WAN interface (FastEthernet0/0) — untrusted external network

Zone pairs were configured in both directions (IN-TO-OUT and OUT-TO-IN) with a service policy allowing TCP, UDP, and ICMP traffic. This ensures that legitimate traffic flows freely while unauthorized access is blocked.

Verification:

```
HQ-Router#
HQ-Router#
HQ-Router#en
HQ-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#
HQ-Router(config)#class-map type inspect match-any ALLOW-TRAFFIC
HQ-Router(config-cmap)# match protocol tcp
HQ-Router(config-cmap)# match protocol udp
HQ-Router(config-cmap)# match protocol icmp
HQ-Router(config-cmap)#
HQ-Router(config-cmap)#policy-map type inspect LAN-TO-WAN-POLICY
HQ-Router(config-pmap)# class type inspect ALLOW-TRAFFIC
HQ-Router(config-pmap-c)# inspect
HQ-Router(config-pmap-c)#
HQ-Router(config-pmap-c)#zone security INSIDE
HQ-Router(config-sec-zone)#zone security OUTSIDE
HQ-Router(config-sec-zone)#
HQ-Router(config-sec-zone)#$T-TO-OUT source INSIDE destination OUTSIDE
HQ-Router(config-sec-zone-pair)#$lcy type inspect LAN-TO-WAN-POLICY
HQ-Router(config-sec-zone-pair)#
HQ-Router(config-sec-zone-pair)#$T-TO-IN source OUTSIDE destination INSIDE
HQ-Router(config-sec-zone-pair)#$lcy type inspect LAN-TO-WAN-POLICY
HQ-Router(config-sec-zone-pair)#
HQ-Router(config-sec-zone-pair)#interface FastEthernet0/1
HQ-Router(config-if)# zone-member security INSIDE
HQ-Router(config-if)#
HQ-Router(config-if)#interface FastEthernet0/0
HQ-Router(config-if)# zone-member security OUTSIDE
HQ-Router(config-if)#
HQ-Router(config-if)#end
HQ-Router#write memory
*Mar 1 00:03:33.799: %SYS-5-CONFIG_I: Configured from console by console
HQ-Router#write memory
Building configuration...
[OK]
HQ-Router#
```

```
HQ-Router#
HQ-Router#show zone security
zone self
    Description: System defined zone

zone INSIDE
    Member Interfaces:
        FastEthernet0/1

zone OUTSIDE
    Member Interfaces:
        FastEthernet0/0

HQ-Router#show zone-pair security
Zone-pair name IN-TO-OUT
    Source-Zone INSIDE Destination-Zone OUTSIDE
    service-policy LAN-TO-WAN-POLICY
Zone-pair name OUT-TO-IN
    Source-Zone OUTSIDE Destination-Zone INSIDE
    service-policy LAN-TO-WAN-POLICY

HQ-Router#ping 192.168.2.1 source 192.168.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/147/224 ms
HQ-Router#ping 192.168.3.1 source 192.168.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/92/128 ms
HQ-Router#
```

7. AAA + SSH Configuration

AAA (Authentication, Authorization, and Accounting) was configured on all routers using the local user database. SSH Version 2 was enabled for secure remote management, and Telnet was disabled to prevent insecure access.

Configuration Details:

- Username: admin (Privilege Level 15)
- Enable Secret: cisco123
- SSH Version: 2.0
- RSA Key Size: 1024 bits
- Domain Name: cisco.com
- VTY Lines: SSH only (Telnet disabled)

Verification:

```
Branch-B-Router#
Branch-B-Router#en
Branch-B-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Branch-B-Router(config)#
Branch-B-Router(config)#username admin privilege 15 secret cisco123
Branch-B-Router(config)#enable secret cisco123
Branch-B-Router(config)#
Branch-B-Router(config)#aaa new-model
Branch-B-Router(config)#aaa authentication login default local
Branch-B-Router(config)#aaa authorization exec default local
Branch-B-Router(config)#
Branch-B-Router(config)#ip domain-name cisco.com
Branch-B-Router(config)#crypto key generate rsa modulus 1024
The name for the keys will be: Branch-B-Router.cisco.com

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Branch-B-Router(config)#ip ssh version 2
Branch-B-Router(config)#
Branch-B-Router(config)#line vty 0 4
Branch-B-Router(config-line)# transport input ssh
Branch-B-Router(config-line)# login authentication default
Branch-B-Router(config-line)#
Branch-B-Router(config-line)#line console 0
Branch-B-Router(config-line)# login authentication default
Branch-B-Router(config-line)#
Branch-B-Router(config-line)#end
Branch-B-Router#write memory
*Mar  1 00:09:58.343: %SSH-5-ENABLED: SSH 1.99 has been enabled
*Mar  1 00:09:59.011: %SYS-5-CONFIG_I: Configured from console by console
Branch-B-Router#write memory
Building configuration...
[OK]
Branch-B-Router#
```

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```
HQ-Router#
HQ-Router#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
HQ-Router#
HQ-Router#ssh -l admin 192.168.2.1

Password:

Branch-A-Router#
```

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8. Device Hardening

Device hardening measures were implemented on all routers to minimize attack surface and enhance security:

- CDP disabled (no cdp run) — prevents network topology disclosure
- HTTP/HTTPS server disabled — prevents web-based attacks
- Unnecessary services disabled (finger, tcp/udp small servers)
- Password encryption enabled (service password-encryption)
- Minimum password length: 8 characters

- Login banner (MOTD) configured — warns unauthorized users
- Console and VTY timeout: 5 minutes

Verification:

```
Branch-A-Router#en
Branch-A-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Branch-A-Router(config)#
Branch-A-Router(config)#no cdp run
Branch-A-Router(config)#no ip http server
Branch-A-Router(config)#no ip http secure-server
Branch-A-Router(config)#no service finger
Branch-A-Router(config)#no service tcp-small-servers
Branch-A-Router(config)#no service udp-small-servers
Branch-A-Router(config)#service password-encryption
Branch-A-Router(config)#security passwords min-length 8
Branch-A-Router(config)#
Branch-A-Router(config)#banner motd #
Enter TEXT message. End with the character '#'.
=====
  AUTHORIZED ACCESS ONLY - Branch-A-Router
  Unauthorized access is strictly prohibited
=====
#
Branch-A-Router(config)#
Branch-A-Router(config)#line console 0
Branch-A-Router(config-line)# exec-timeout 5 0
Branch-A-Router(config-line)#
Branch-A-Router(config-line)#line vty 0 4
Branch-A-Router(config-line)# exec-timeout 5 0
Branch-A-Router(config-line)#
Branch-A-Router(config-line)#end
Branch-A-Router#write memory
*Mar 1 00:00:53.263: %SYS-5-CONFIG_I: Configured from console by admin on console
Branch-A-Router#write memory
Building configuration...
[OK]
Branch-A-Router#
*Mar 1 00:01:03.911: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0 from LOADING to FULL, L
Branch-A-Router#
```

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```
=====
  AUTHORIZED ACCESS ONLY - HQ-Router
  Unauthorized access is strictly prohibited
=====

User Access Verification
Username: admin
Password:
HQ-Router#
*Mar 1 00:01:06.631: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0 from LOADING to FULL, Loading Done
HQ-Router#
```

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9. Syslog / NTP / SNMP

Centralized monitoring and logging were configured on all routers:

Syslog:

- Syslog server: 192.168.1.10 (HQ-PC1)
- Logging level: Informational

NTP:

- NTP server: 192.168.1.10 (HQ-PC1)
- Ensures time synchronization across all devices

SNMP:

- Community string: public (Read Only)
- Community string: private (Read Write)
- SNMP traps enabled for all events

```
Branch-B-Router#
Branch-B-Router#en
Branch-B-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Branch-B-Router(config)#
Branch-B-Router(config)#logging 192.168.1.10
Branch-B-Router(config)#logging trap informational
Branch-B-Router(config)#logging on
Branch-B-Router(config)#
Branch-B-Router(config)#ntp server 192.168.1.10
Branch-B-Router(config)#
Branch-B-Router(config)#snmp-server community public RO
Branch-B-Router(config)#snmp-server community private RW
Branch-B-Router(config)#snmp-server location BranchB-Site
Branch-B-Router(config)#snmp-server contact admin@cisco.com
Branch-B-Router(config)#snmp-server enable traps
% Cannot enable both sham-link state-change interface traps.
% New sham link interface trap not enabled.
Branch-B-Router(config)#
Branch-B-Router(config)#end
Branch-B-Router#write memory
*Mar 1 00:05:29.391: %SYS-5-CONFIG_I: Configured from console by admin on console
Branch-B-Router#write memory
*Mar 1 00:05:30.395: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.10 port 514 started - CLI
initiated
Branch-B-Router#write memory
Building configuration...
[OK]
Branch-B-Router#
```

10. PC Ips:

HQ- PCs Ips:

```
HQ-PC1>
HQ-PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

HQ-PC1> save
Saving startup configuration to startup.vpc
. done

HQ-PC1> show ip

NAME       : HQ-PC1[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10033
RHOST:PORT : 127.0.0.1:10034
MTU       : 1500

HQ-PC1>
```

```
HQ-PC2> ip 192.168.1.11 255.255.255.0 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

HQ-PC2> save
Saving startup configuration to startup.vpc
. done

HQ-PC2> █
```

Branch-A-PCs Ips:

```
Branch-A-PC1> ip 192.168.2.10 255.255.255.0 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1

Branch-A-PC1> save
Saving startup configuration to startup.vpc
. done

Branch-A-PC1> █
```

```
Branch-A-PC2> ip 192.168.2.11 255.255.255.0 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.11 255.255.255.0 gateway 192.168.2.1

Branch-A-PC2> save
Saving startup configuration to startup.vpc
. done

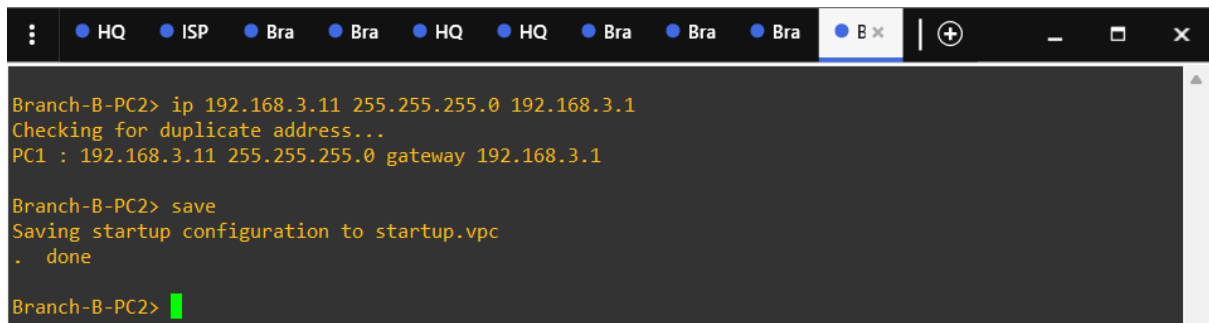
Branch-A-PC2> █
```

Branch-B-PCs Ips:

```
Branch-B-PC1> ip 192.168.3.10 255.255.255.0 192.168.3.1
Checking for duplicate address...
PC1 : 192.168.3.10 255.255.255.0 gateway 192.168.3.1

Branch-B-PC1> save
Saving startup configuration to startup.vpc
. done

Branch-B-PC1> █
```

A screenshot of a GNS3 terminal window. The title bar shows several tabs labeled 'HQ', 'ISP', 'Bra', and 'B x'. The terminal content shows the following commands and output:

```
Branch-B-PC2> ip 192.168.3.11 255.255.255.0 192.168.3.1
Checking for duplicate address...
PC1 : 192.168.3.11 255.255.255.0 gateway 192.168.3.1

Branch-B-PC2> save
Saving startup configuration to startup.vpc
. done

Branch-B-PC2>
```

11. Troubleshooting

During implementation, the following issues were encountered and resolved:

Issue 1: Router Hanging/Freezing

Problem: c3725 routers were hanging during operation. Solution: Increased RAM from 128MB to 256MB and verified idlepc value (0x60c09aa0) was correctly set.

Issue 2: IPsec VPN Tunnel Not Establishing

Problem: VPN tunnel between HQ and Branch-A was not establishing. Solution: OSPF adjacency had not fully formed after router reboot. After OSPF reached FULL state, the tunnel was triggered using source ping and established successfully.

Issue 3: PC-to-PC Ping Timeout

Problem: First 1-2 packets timed out during PC-to-PC ping. Solution: This is normal behavior — the first packet triggers the IPsec VPN tunnel establishment. Subsequent packets succeed after the tunnel is up.

Issue 4: ISP Router Misconfiguration

Problem: IPsec VPN was accidentally configured on ISP-Router. Solution: All crypto configurations were removed from ISP-Router using "no" commands as ISP-Router is only a transit device.

12. Challenges Faced

The following challenges were encountered during this assignment:

- **Router Performance:** The c3725 routers consumed significant CPU resources in GNS3, causing occasional freezing. This was resolved by increasing RAM and ensuring correct idlepc values.
- **IOS Compatibility:** The initial c7200 IOS image did not support Zone-Based Firewall commands, requiring a switch back to c3725 with the adventerprise9 image.
- **VPN Tunnel Timing:** IPsec tunnels require traffic to trigger establishment, which caused initial confusion when pings failed before the tunnel was up.

- ZBF and VPN Integration: Ensuring ZBF policies allowed VPN-encrypted traffic while still blocking unauthorized access required careful policy configuration.

13. Conclusion

This assignment successfully demonstrated the implementation of a complete enterprise network security solution using industry-standard technologies. The following objectives were achieved:

- ✓ Secure branch-to-branch communication using IPsec VPN tunnels
- ✓ Network traffic protection using Zone-Based Firewall policies
- ✓ Secure device management using AAA authentication and SSH Version 2
- ✓ Device hardening to minimize attack surface
- ✓ Centralized monitoring using Syslog, NTP, and SNMP
- ✓ Dynamic routing using OSPF with proper security configurations

The implementation provided hands-on experience with real-world enterprise security technologies and best practices. The troubleshooting process enhanced problem-solving skills and deepened understanding of network security concepts.